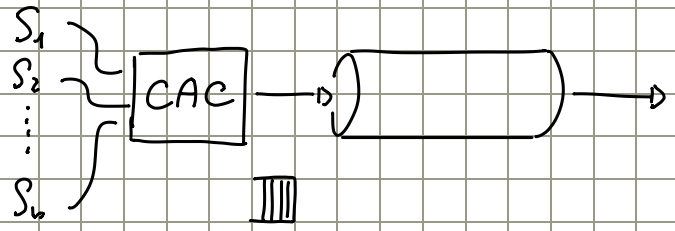
30/10/2026



TBF → p_i: Committed Information Rate (CIR)
→ b_i: Burst Size

$$\sum_{i=1}^{N} p_i < C$$

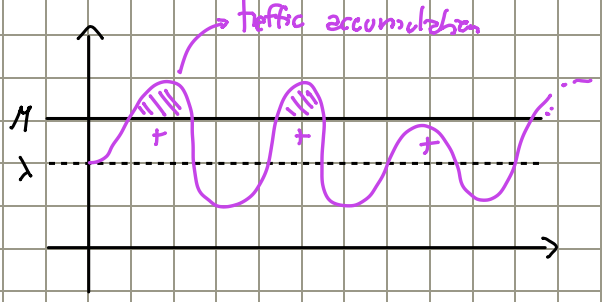
$$\Rightarrow \text{Aggregate traffic is sustainable}$$

$$\sum_{i=1}^{N} b_i < B$$

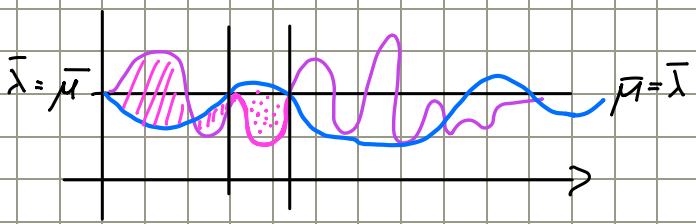


if $\mu > \lambda$ → queue is stable

$\boxed{\mu = \lambda}$ → queue is undetermined



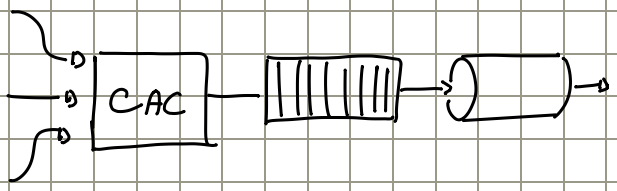
we have much pckt. to consume the traffic rather than accumulate backlog



$$\begin{cases} \sum_i p \le C \\ \sum_i b_i \le B \end{cases} \longrightarrow$$

Stochastic CAC

$\mathcal{E}$ = prob. of cell blocking



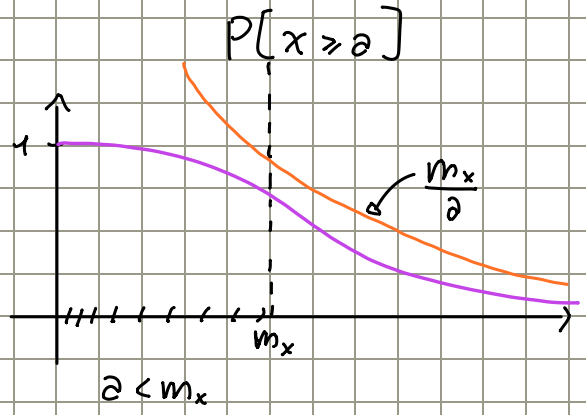
How many sources can be admitted into the system while limiting the prob. of buffer overflowing to $\mathcal{E}$.

$\mathcal{E} = 10^{-6} \rightarrow$ overflow prob. is $\le 5 \cdot 10^{-7}$

Recall [Main Bounds of Statistical Analysis]

- Markov inequality: $\forall\, x \ge 0;\ \forall\, a \in \mathbb{R}_+$

$$\mathbb{P}[x \ge a] \le \frac{m_x}{a}$$



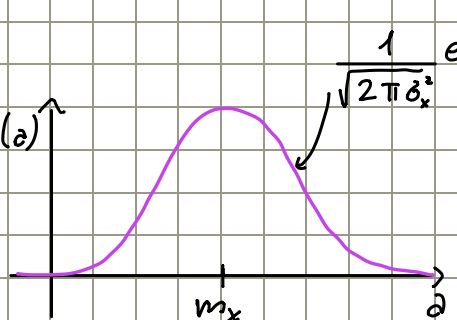
Ex:

$$x \sim \mathcal{E}(\lambda) \rightarrow \mathbb{P}\left[x \ge 4 \cdot \frac{1}{\lambda}\right] \le \frac{\frac{1}{\lambda}}{4 \cdot \frac{1}{\lambda}} = 0,25$$

$$e^{-\lambda\left(4 \cdot \frac{1}{\lambda}\right)} = e^{-4} \ll 0,25$$

$$\mathbb{P}[x > 4 m_x] \overset{?}{\le}$$

$$x \sim N(m_x, \sigma_x^2)$$

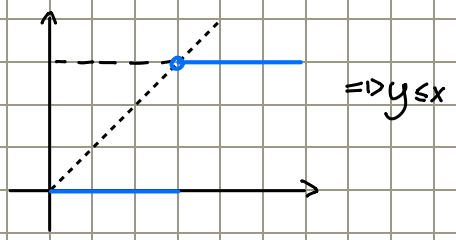


If $F_x(0) = P[x<0] \ll P[x > m_x \cdot 4] \leq 0,25$

Proof

$X \geq 0$

$$y \triangleq \begin{cases} 0 & \text{if } x < a \\ a & \text{if } x \geq a \end{cases}$$



$\Rightarrow y \leq x$

$y \triangleq a \, \chi\{x \geq a\}$

↑ indicator function $\quad \chi\{A\} = \begin{cases} 1 & \text{if } A \text{ is true} \\ 0 & \text{otherwise} \end{cases}$

$y \leq x \Rightarrow y - x \leq 0 \Rightarrow E[y-x] \leq 0$

$E[y-x] =$

$E[y-x] = \underbrace{E[y]}_{a\,E[\chi\{x \geq a\}]} - \underbrace{E[x]}_{m_x} \leq 0$

$a\,P[x \geq a]$

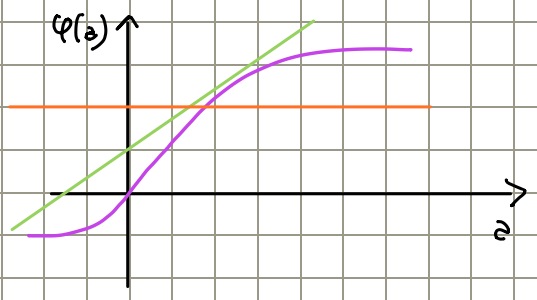
$a\,P[x \geq a] - m_x \cdot 0$

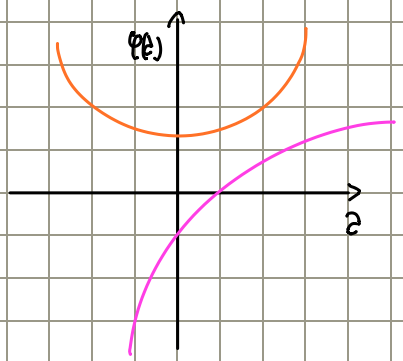
$P[x \geq a] \leq \frac{m_x}{a}$

QED

Corollary

$\forall x, \quad \forall \varphi : \mathbb{R} \to \mathbb{R}$ s.t. $\begin{cases} \varphi'(a) \geq 0 & \text{(must not be decreasing)} \\ \varphi(0) \geq 0 & \text{(non negative in positive axis)} \end{cases}$





$$P[|x| \geq a] \leq \frac{E[\varphi(|x|)]}{\varphi(a)}$$

Proof

$$P[|x| \geq a] = P[\underbrace{\varphi(|x|)}_{\geq 0} \geq \varphi(a)] \leq \frac{E[\varphi(|x|)]}{\varphi(a)}$$

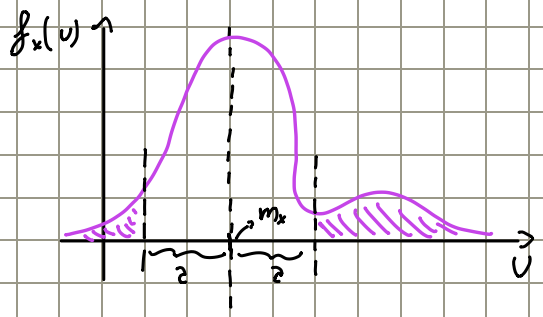
↑ Since $|x| \geq 0$ and $\varphi(a)$ is non decr. and non neg. for pos. values

↑ $\varphi(|x|) \geq 0 \rightarrow$ Markov Bound

Chebyshev Bound

$\forall_X: \; E[X] < \infty \qquad E[X^2] < \infty$

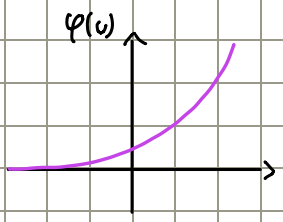
$\rightarrow P[|X - m_X| > a] \leq \frac{\sigma_X^2}{a^2}$



Proof:

$Y \triangleq X - m_X$

$\varphi(v) = \begin{cases} 0 & \text{if} \quad v < 0 \\ v^2 & \text{if} \quad v \geq 0 \end{cases}$



$P[|Y| > a] \leq \frac{E[\varphi(|Y|)]}{\varphi(a)} =$

(Markov)

$X \sim \mathcal{E}(\lambda)$

$E[X] = 1/\lambda$

$P[X > \underbrace{4 m_X}_{a}]$ → Markov ineq. $\leq \frac{m_X}{a} = \frac{1}{4} = 0{,}25$

e^{-4}

$P[|X - m_X| > \tilde{a}] = P[X - m_X > \tilde{a} \cup -X + m_X > \tilde{a}]$

$\tilde{a} > 0$

$= P[X - m_X > \tilde{a}] + P[X - m_X < -\tilde{a}]$

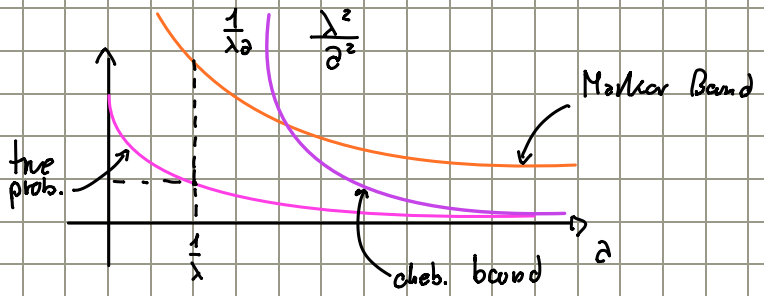
$= P[X \geq 4 m_X] + P[X \leq -2 m_X] = P[X > m_X 4] \leq \frac{\sigma_X^2}{\tilde{a}^2} = \frac{1}{9}$

$\sigma_X^2 = \frac{1}{\lambda^2}$

$\tilde{a} = 3 m_X = 3 \frac{1}{\lambda}$

$\tilde{a} = 3 m_X > 0$

$$P[X \geq a] = e^{-\lambda a}$$



Question by prof

$\forall x,\ E[x] < \infty$

$E[x^2] < \infty$

$E[(x-m_x)^2] < \infty$ (σ_x^2)

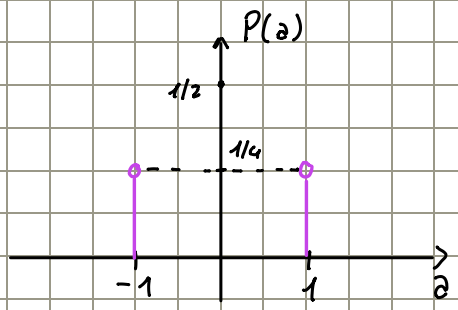
$$\sigma_x^2 = E[x^2] - E[x]^2$$

$$\exists x \text{ s.t. } P[|x-m_x| \geq a] = \frac{\sigma_x^2}{a^2}$$

(strict)

$x \in \{-1, 0, 1\}$; $P(1) = P(-1) = 1/4$

$P(0) = 1/2$



$$E[x] = 0$$

$$= -\frac{1}{4} + 0 \cdot \frac{1}{2} + \frac{1}{4}$$

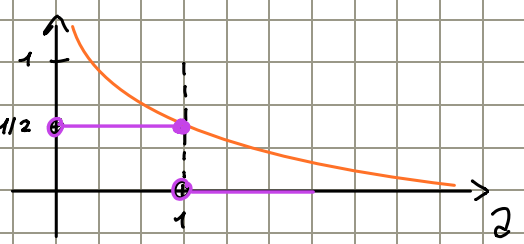
$$\sigma_x^2 = E[x^2] = (-1)^2 \cdot 1/4 + 0 \cdot \frac{1}{2} + 1^2 \cdot \frac{1}{4} = \frac{1}{2}$$

$$P[|x - m_x| \geq a] = P[|x| \geq a] \leq \frac{\sigma_x^2}{a^2} = \frac{1}{4a^2}$$

($m_x = 0$)

$a = 1$ $\quad P[|x| \geq 1] \leq 1/2$

$= \frac{1}{2}$



Chernoff Band

$P[\text{buffer overflow}] \leq \varepsilon$

$\leq \frac{\partial x^2}{\partial^2} \leq \varepsilon$